

EG1311: Design & Make

Project Materials

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Project: Materials

- Teams receive project materials in first Tutorial
 - Electronics Kit (x2)
 - Tools Kit
 - Materials Kit
- Only parts provided should be used
- Exception: glue/tape, paper, & paper clips
- Teams responsible for keeping materials well
- Some items used. If broken, ask for exchange

Project: Tools Kit

Tools Kit	Qty
Universal Battery Charger	1
9V Rechargeable Battery	2
AA Rechargeable Battery	12
Hot Glue Gun	1
A4 Cutting Mat	1
Retractable Pen Knife	1
30cm Steel Ruler	1
Ping Pong Ball	1

Project: Materials Kit

Materials Kit	Qty
(★) 50cm × 50cm 2mm Cardboard	1
(★) A4 2mm Polypropylene	1
(★) A4 5mm Foamcore	1
(★) Anti-slip mat sheet	1
(★) Roll of masking tape	1
(★) Popsicle Sticks	20
(★) 3” diameter rubber bands	10
(★) Glue Sticks	3

Cardboard

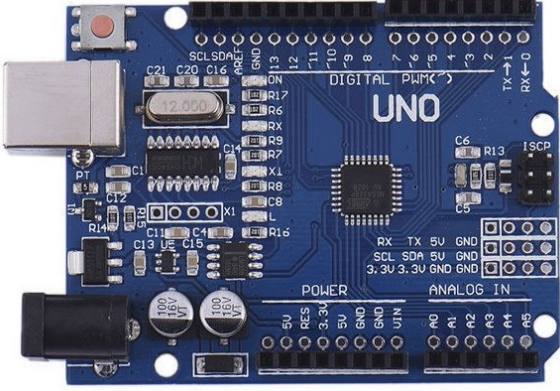
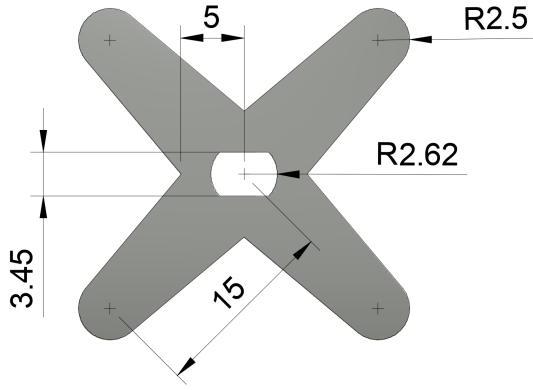
- Corrugation Grain
- Cut against grain first
- Careful of pen knife triangular blade
- Print templates at 100% scale
- Tape on template and cut both layers



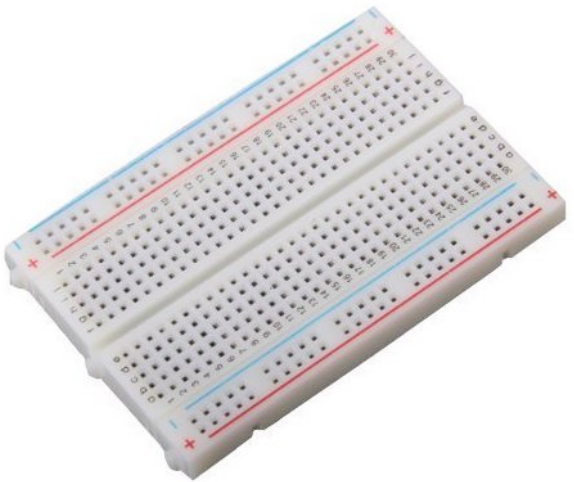
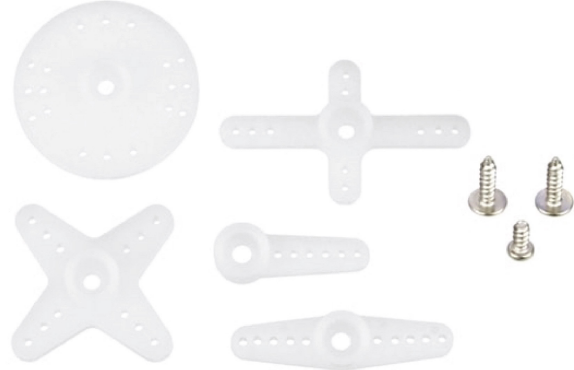
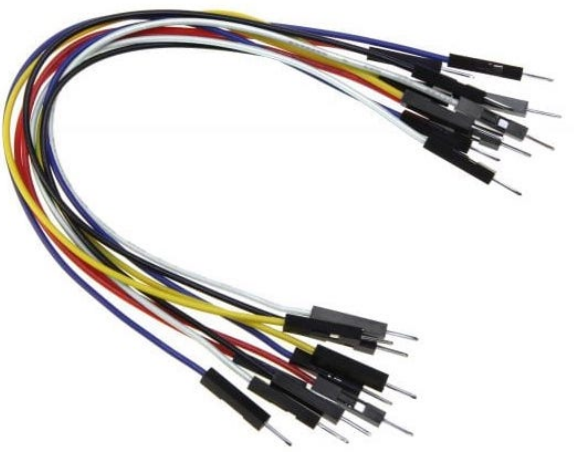
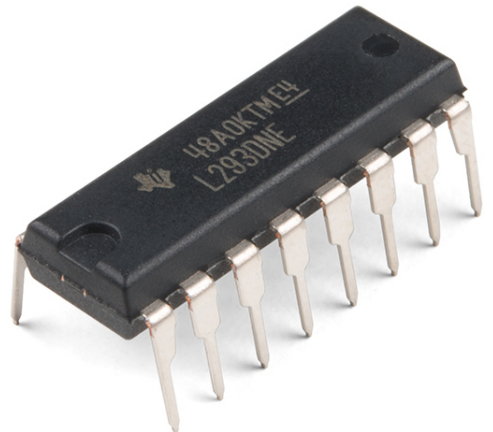
Acrylic

- Laser cut 3mm black acrylic
- Cutting allowed starting from Thursday

Project: Electronics Kit (x2)

Image	Part	Image	Part
	Arduino Uno Qty 1 Datasheet		USB Cable Qty 1 USB-B to USB-A
	Geared Motor Qty 2 120:1 gear ratio 3-9V volts CAD Drawing		9V Battery Holder Qty 1
	Motor Coupler Qty 2		6 x AA Battery Holder Qty 1

Project: Electronics Kit (x2)

Image	Part	Image	Part
	Servo Motor (SG90) Qty 1 Datasheet		Breadboard 400-Tie-Point Qty 1 How to use
	Servo Motor Coupler Pack Qty 1		(★) 40cm Jumper Wires Qty 20 Male-Male Qty 10 Male-Female
	Ultrasonic Sensor (HC-SR04) Qty 1 Datasheet		H-Bridge (L293DNE) Motor Driver Qty 1 Datasheet

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